

Erjona Topalli

Senior Software Engineer | Frontend Platform Engineering |
React • TypeScript • Preact • Performance Optimization
MIT | Oracle

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EXPERIENCE

Oracle — Senior Software Engineer

JULY 2017- PRESENT

Led modernization efforts for Oracle JET's visualization components, migrating legacy implementations to Preact while maintaining backward compatibility for a frontend toolkit used by 6,000+ developers across Oracle.

Owned the design and implementation of modernized visualization components, establishing reusable patterns that improved maintainability, consistency, and long-term scalability.

Improved rendering performance of core visualization components by up to 45% through targeted rendering optimizations and component redesign.

Researched and prototyped performance improvements—including replacing legacy SVG rendering with native HTML where appropriate—to reduce rendering overhead and improve responsiveness.

Defined stable component APIs and migration strategies that enabled incremental adoption of modern implementations without disrupting existing Oracle applications.

Partnered with UX and accessibility specialists to deliver enterprise-grade visualization components while maintaining WCAG compliance.

Facilitated architecture reviews, mentored engineers through code reviews and design discussions, and promoted engineering best practices across the Oracle JET platform.

Leveraged AI-assisted development tools to accelerate prototyping, implementation, code generation, and technical research.

PROJECTS

Rock Climbing Gym Web App, Oaktown Boulders — Founding Engineer 2020-2022

Designed and implemented a full-stack route management platform.
Built an interactive wall editor for staff to create and modify climbing routes.

SKILLS

Languages

TypeScript • JavaScript •
Python • Java

Frontend

React • Preact • HTML5 • CSS3 •
D3.js

Backend

Node.js • Express • REST APIs •
MongoDB

Cloud & Tools

AWS • Git • Cursor • Figma

Expertise Areas

Frontend Platform
Engineering • Component
Architecture • Performance
Optimization • Accessibility
(WCAG) • API Design •
Developer Experience •
Technical Leadership

EDUCATION

Massachusetts Institute of Technology (MIT) GPA 4.2 / 5.0

BSc Computer Science,
BSc Mathematics